

The Regularity Lemma for Stable Graphs and its Applications in Property Testing

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Introduction to Szemerédi's Regularity Lemma

The idea of the SzRL is that all graphs can be partitioned in such a way that most pairs of parts behave in a quasi-random way, a property which is formalized as *regularity*.

Definition

Given $\epsilon > 0$ and a graph G , a pair of (not necessarily disjoint) subsets of vertices $A, B \subseteq G$ is said to be ϵ -regular if for all $A' \subseteq A$ and $B' \subseteq B$ such that $|A'| \geq \epsilon|A|$ and $|B'| \geq \epsilon|B|$, we have

$$|d(A', B') - d(A, B)| \leq \epsilon.$$

Introduction to Szemerédi's Regularity Lemma

When most of the pairs satisfy this property, we say that the partition is *regular*.

Definition

Let G be a graph and let $\epsilon > 0$. An ϵ -regular partition of G is a partition of its vertex set into k parts $\{A_1, \dots, A_k\}$ with remainder set B such that:

- $|B| \leq \epsilon|G|$, and may be empty.
- All but at most ϵk^2 of the pairs (A_i, A_j) with $1 \leq i < j \leq k$ are ϵ -regular.

Introduction to Szemerédi's Regularity Lemma

Theorem (Szemerédi's Regularity Lemma, [?])

For every $\epsilon > 0$ and every positive integer m , there exists a positive integer $M = M(\epsilon, m)$ such that every graph with at least m vertices admits an ϵ -regular partition $\{A_1, \dots, A_k\}$ and remainder B with $m \leq k \leq M$, with $|A_1| = \dots = |A_k|$.

Crucially, the theorem proves that the number of parts k does not grow with the size of the graph, but only depends on ϵ (and m).

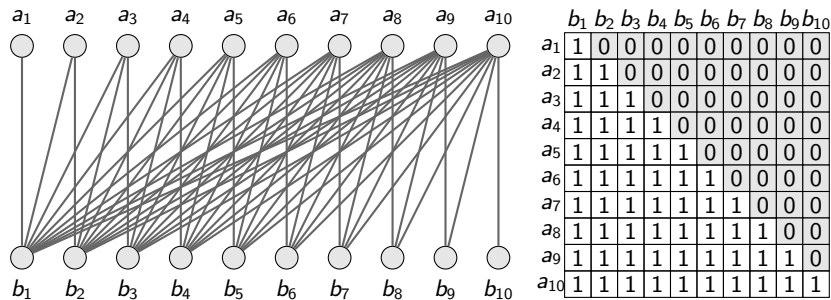
Introduction to Szemerédi's Regularity Lemma

However, the lemma has two major drawbacks:

- 1 The bound on the number of parts M is a *tower-type* function of ϵ , i.e. a power-tower of height polynomial in $1/\epsilon$. This is unavoidable, as shown by Gowers [?].
- 2 The partition may contain some irregular pairs. This is also unavoidable, and the half-graph H_n is an example.

Stable Graphs and the Stable Regularity Lemma

Stable graphs are those that do not contain a *half-graph* as a bi-induced subgraph.



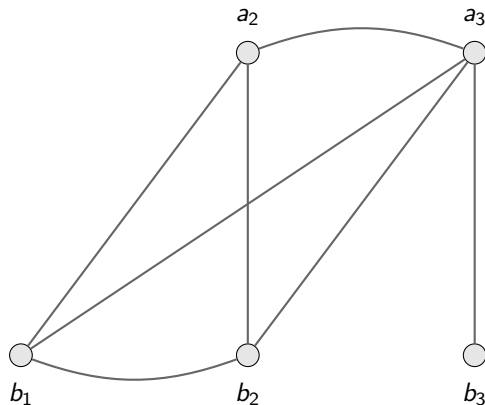
Stable Graphs and the Stable Regularity Lemma

Definition

Let G be a graph. We say that G has the *k -order property* if there exist two sequences of vertices $\langle a_i \mid i \in \{1, \dots, k\} \rangle$ and $\langle b_i \mid i \in \{1, \dots, k\} \rangle$ such that for all $i, j \leq k$, $a_i R b_j$ if and only if $i \geq j$. Otherwise, we say that G has the *non- k -order property* or that G is *k -stable*.

The two sequences need not to be disjoint, and any combination of edges and non-edges may appear within the vertices of each sequence.

Stable Graphs and the Stable Regularity Lemma



	a_2	a_3	b_1	b_2	b_3
a_2	0	1	1	1	0
a_3	1	0	1	1	1
b_1	1	1	0	1	0
b_2	1	1	1	0	0
b_3	0	1	0	0	0

Stable Graphs and the Stable Regularity Lemma

Theorem (Stable Regularity Lemma, [?])

Each k -stable graph G admits an equitable partition of its vertex set with at most $c(k) \cdot (1/\epsilon)^{2^{k+1}}$ parts such that ALL pairs of parts are ϵ -regular and have density either less than ϵ or greater than $1 - \epsilon$.

Notably both limitations of the SzRL are overcome: the bound on the number of parts is polynomial in $1/\epsilon$, and there are no irregular pairs.

Sketch of the Proof of the Stable Regularity Lemma

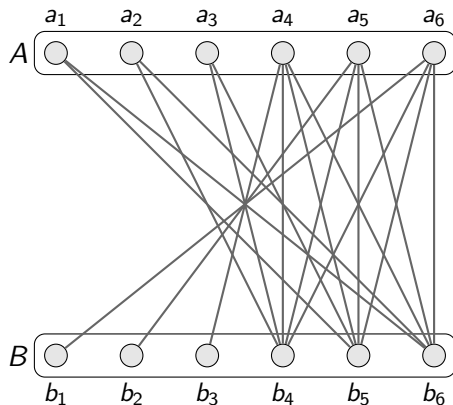
Definition (Good, [?])

Let G be a finite graph with the non- k_* -property. We say that $A \subseteq G$ is ϵ -good when for every $b \in G$ the truth value $t = t(b, A) \in \{0, 1\}$ satisfies $|\overline{B}_{A,b}| = |\{a \in A \mid aRb \neq t\}| < \epsilon|A|$.

Definition (Excellent, [?])

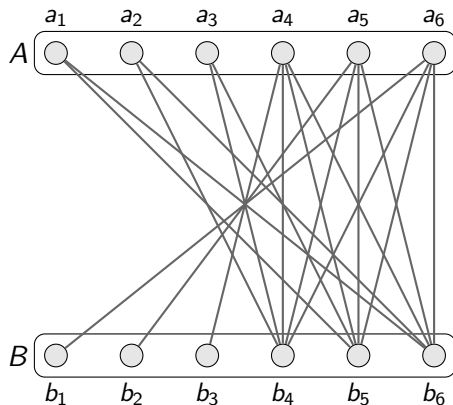
Let G be a finite graph with the non- k_* -property. We say that $A \subseteq G$ is (ϵ, ζ) -excellent when A is ϵ -good and, if B is ζ -good, then the truth value $t = t(B, A)$ satisfies $|\{a \in A \mid t(a, B) \neq t(A, B)\}| < \epsilon|A|$. In particular, we say A is ϵ -excellent if A is (ϵ, ϵ) -excellent.

Sketch of the Proof of the Stable Regularity Lemma



	b_1	b_2	b_3	b_4	b_5	b_6
a_1	0	0	0	0	1	1
a_2	0	0	0	1	0	1
a_3	0	0	0	1	1	0
a_4	0	0	1	1	1	1
a_5	0	1	0	1	1	1
a_6	1	0	0	1	1	1

Sketch of the Proof of the Stable Regularity Lemma



	b_1	b_2	b_3	b_4	b_5	b_6
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a_6	1	0	0	1	1	1

Do excellent subsets always exist?

Sketch of the Proof of the Stable Regularity Lemma

Definition (k -tree)

A k -tree in G is an ordered pair $H = (\bar{c}, \bar{b})$ comprising two (not necessarily disjoint) sequences:

- $\bar{c} = \{c_\eta \in G \mid \eta \in \{0, 1\}^{<k}\}$, the set of *nodes* and
- $\bar{b} = \{b_\rho \in G \mid \rho \in \{0, 1\}^k\}$, the set of *branches*,

satisfying that for all $\eta \in \{0, 1\}^{<k}$ and $\rho \in \{0, 1\}^k$, if for some $\ell \in \{0, 1\}$ we have $\eta \frown \langle \ell \rangle \triangleleft \rho$, then $b_\rho R c_\eta \equiv \ell$.

Sketch of the Proof of the Stable Regularity Lemma

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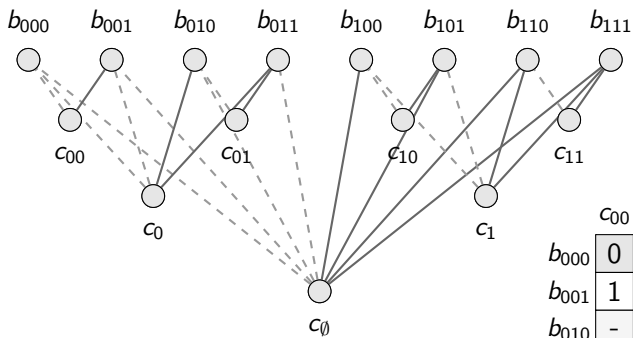
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Definition (Tree bound)

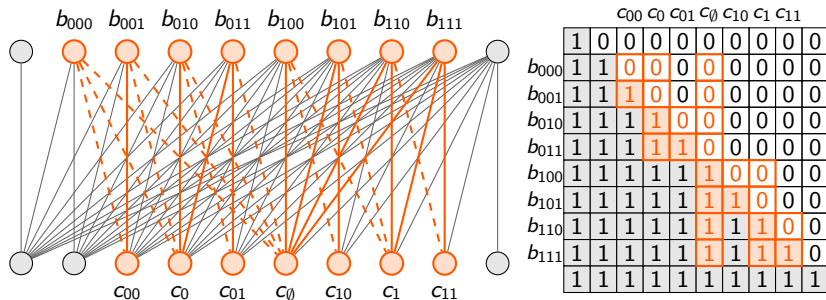
Suppose G is a finite graph. We denote the *tree bound* $k_{**} = k_{**}(G)$ as the minimal positive integer such that there is no k_{**} -tree $H = (\bar{c}, \bar{b})$ in G .

Sketch of the Proof of the Stable Regularity Lemma



	c_{00}	c_0	c_{01}	$c_{\emptyset}$	c_{10}	c_1	c_{11}
b_{000}	0	0	-	0	-	-	-
b_{001}	1	0	-	0	-	-	-
b_{010}	-	1	0	0	-	-	-
b_{011}	-	1	1	0	-	-	-
b_{100}	-	-	-	1	0	0	-
b_{101}	-	-	-	1	1	0	-
b_{110}	-	-	-	1	-	1	0
b_{111}	-	-	-	1	-	1	1

Sketch of the Proof of the Stable Regularity Lemma



Sketch of the Proof of the Stable Regularity Lemma

Theorem (Lemma 6.7.9 in [?])

*If a graph G has the $2^{k_{**}}$ -order property, then the tree bound of G is at least $k_{**} + 1$.*

Sketch of the Proof of the Stable Regularity Lemma

Theorem (Lemma 6.7.9 in [?])

*If a graph G has the $2^{k_{**}}$ -order property, then the tree bound of G is at least $k_{**} + 1$.*

*On the other hand, if a graph G has tree bound at least $k_{**} = 2^{k_*+1} - 1$, then it has the k_* -order property.*

Sketch of the Proof of the Stable Regularity Lemma

Lemma (Existence of excellent subsets, [?])

Let G be a finite graph with the non- k_ -order property. Let $\zeta \leq \frac{1}{2^{k_{**}}}$, $\epsilon \in (0, \frac{1}{2})$. Then, for every $A \subseteq G$ with $|A| \geq \frac{1}{\epsilon^{k_{**}}}$ there exists an (ϵ, ζ) -excellent subset $A' \subseteq A$ such that $|A'| \geq \epsilon^{k_{**}-1}|A|$.*

Sketch of the Proof of the Stable Regularity Lemma

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TO DO

Sketch of the Proof of the Stable Regularity Lemma

$$A_{\emptyset} := A$$

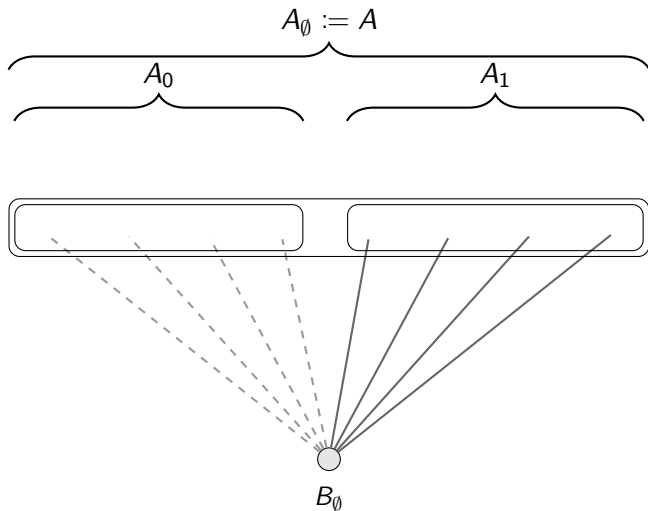

Sketch of the Proof of the Stable Regularity Lemma

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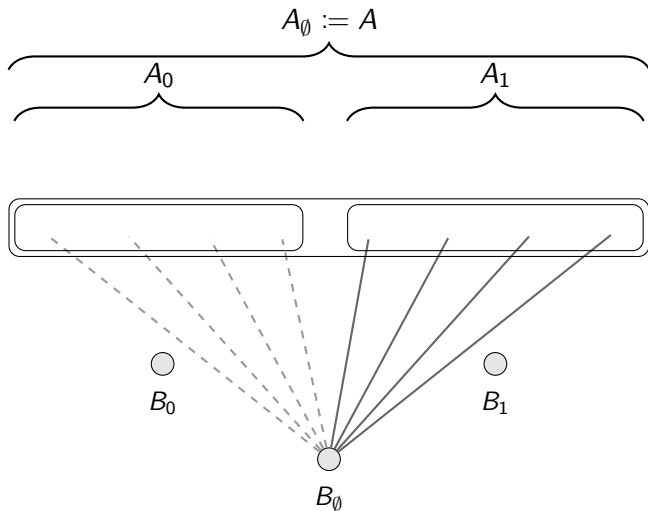



$$B_{\emptyset}$$

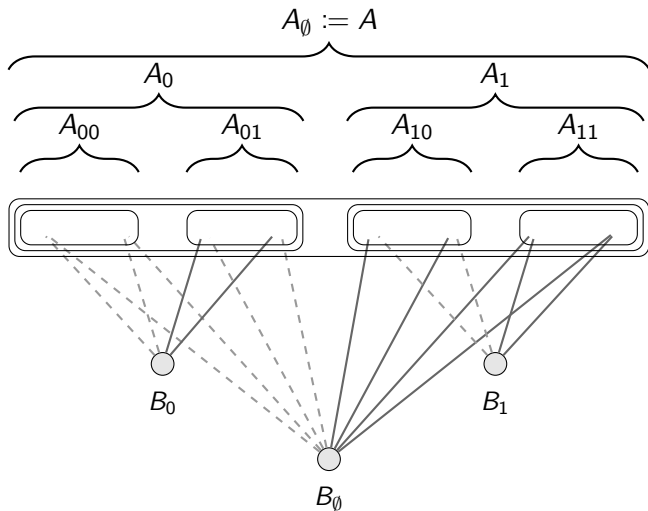
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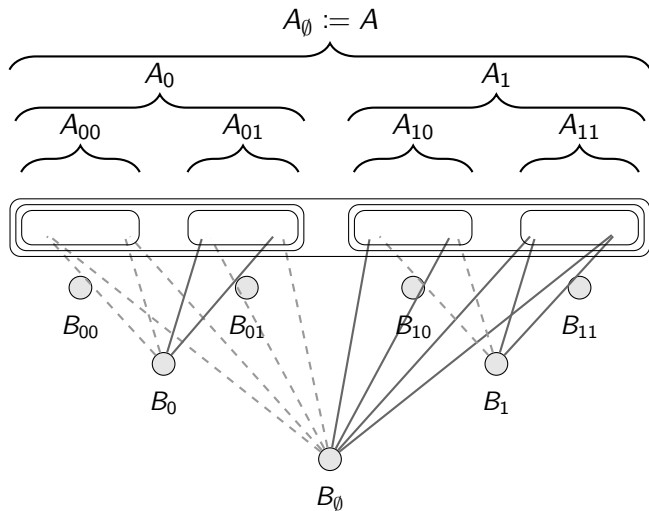
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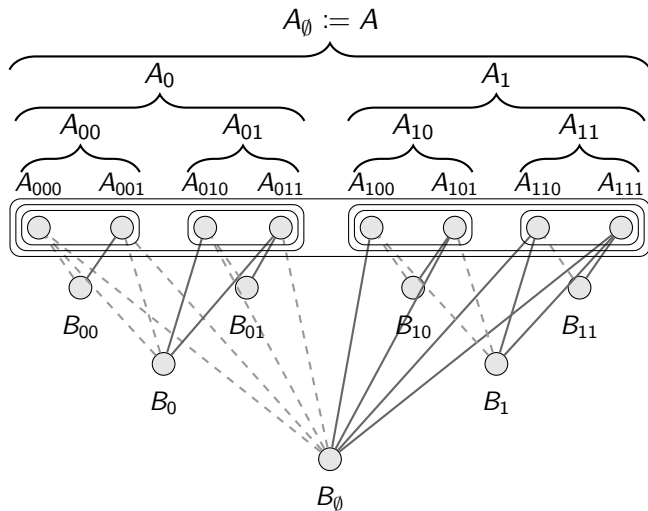
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Sketch of the Proof of the Stable Regularity Lemma

Lemma (Existence of excellent subsets, [?])

Let G be a finite graph with the non- k_ -order property. Let $\zeta < \frac{1}{2^{k_{**}}}$, $\epsilon \in (0, \frac{1}{2})$. Let $\langle m_\ell \mid \ell \in \{0, \dots, k_{**}\} \rangle$ be a decreasing sequence of natural numbers such that $\epsilon m_\ell \geq m_{\ell+1}$ for all $\ell \in \{0, \dots, k_{**} - 1\}$ and $m_{k_{**}} \geq 1$. Then, for every $A \subseteq G$ with $|A| \geq m_0$ there exists $(\frac{m_{\ell+1}}{m_\ell}, \zeta)$ -excellent subset $A' \subseteq A$ such that $|A'| = m_\ell$ for some $\ell \in \{0, \dots, k_{**} - 1\}$.*

Sketch of the Proof of the Stable Regularity Lemma

Lemma

Let G be a finite graph with the non- k_* -order property. Let $\epsilon \in (0, \frac{1}{2})$ and $\epsilon' \leq \frac{1}{2^{k_{**}}}$. Let $A \subseteq G$ such that $|A| = n \geq m_0$. Let $\langle m_\ell \mid \ell \in \{0, \dots, k_{**}\} \rangle$ be a decreasing sequence of natural numbers such that $\epsilon m_\ell \geq m_{\ell+1}$ for all $\ell \in \{0, \dots, k_{**} - 1\}$ and $m_{k_{**}} \geq 1$. Denote $m_* := m_0$ and $m_{**} := m_{k_{**}}$. Then, there is a partition $\bar{A} = \langle A_j \mid j \in \{1, \dots, j(*)\} \rangle$ with remainder $B = A \setminus \bigcup_{j < j(*)} A_j$ such that:

- ① For all $j \in \{1, \dots, j(*)\}$, $|A_j| \in \langle m_\ell \mid \ell \in \{0, \dots, k_{**} - 1\} \rangle$.
- ② For all $i \neq j \in \{1, \dots, j(*)\}$, $A_i \cap A_j = \emptyset$.
- ③ For all $j \in \{1, \dots, j(*)\}$, A_j is (ϵ, ϵ') -excellent.
- ④ $|B| < m_*$.

Sketch of the Proof of the Stable Regularity Lemma

Question: Are subsets of excellent sets still excellent?

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Answer: With high probability, yes, but with a small loss in excellence.

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Lemma

Let G be a finite graph with the non- k_ -order property. Then, for every $\epsilon \in (0, \frac{1}{2})$, $\zeta \in (0, \frac{1}{2} - \epsilon)$, $\xi \in (0, 1)$ and $m \geq \frac{1}{\zeta^2}(k_* \log \frac{1}{\zeta^2} k_* - \log \xi)$, if $A \subseteq G$ is an ϵ -good subset of size $n \geq m$, then a subset $A' \subseteq A$ of size m , sampled uniformly at random, is $(\epsilon + \zeta)$ -good with probability $1 - \xi$.*

Sketch of the Proof of the Stable Regularity Lemma

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Lemma

Let G be a finite graph with the non- k_ -order property. Then, for every $\zeta \in \{0, \frac{1}{2}\}$ and $\zeta' < \zeta$, there is $\epsilon_1 = \epsilon_1(\zeta, \zeta')$ such that for every $\epsilon < \epsilon' \leq \epsilon_1$, if*

- $A \subseteq G$ is (ϵ, ϵ') -excellent and*
- $A' \subseteq A$ is $(\epsilon + \zeta')$ -good,*

then, A' is $(\epsilon + \zeta, \epsilon')$ -excellent.

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Property Testing of H -freeness in Stable Graphs

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Conclusion and Future Work

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Thank you!

You can now rate the presentation. Was it *good*, *excellent* or *exceptional*?